

Goal. Find the sum of a geometric or telescoping series, and apply the n th term test.

0. Fill in the following blanks.

- A **sequence** is a _____ of numbers. A sequence converges if its _____ have a limit.
- A **series** is a _____ of numbers. A series converges if its _____ has a limit.
- The **nth term test** tells us that the series $\sum_{n=1}^{\infty} a_n$ _____ if $\lim_{n=0} a_n \neq$ _____, and if $\lim_{n=0} a_n =$ _____, it tells us _____.

1. A *geometric series* is a series of the form, $\sum_{n=0}^{\infty} ar^n$ for constants a and r . When does such a series converge? If the series does converge, what does it converge to?

2. Find the fractional expansion of the repeating decimal $0.42424242\overline{42}$ by rewriting it as a geometric series.

3. Determine whether the series converges or diverges. If it converges, find the sum.

$$1 - \frac{1}{\sqrt{3}} + \frac{1}{3} - \frac{1}{3\sqrt{3}} + \frac{1}{9} - \frac{1}{9\sqrt{3}} + \cdots$$

4. Determine whether the series $\sum_{n=1}^{\infty} \frac{4}{5n}$ converges or diverges. If it converges, find the sum.

5. Determine whether the series $\sum_{n=5}^{\infty} \left(1 + \frac{1}{n}\right)^n$ converges or diverges. If it converges, find the sum.

6. Determine whether the series $\sum_{n=1}^{\infty} \frac{3}{n(n+2)}$ converges or diverges. If it converges, find the sum. (Hint: Use partial fractions.)

7. Determine whether the series $\sum_{n=1}^{\infty} \ln(n) - \ln(n+1)$ converges or diverges. If it converges, find the sum.

8. Determine whether the series $\sum_{n=0}^{\infty} \frac{7 + e^n}{\pi^n}$ converges or diverges. If it converges, find the sum. (Hint: Use limit laws to split into two series.)